

Empowering ABC Foodmart: *Real-Time Insights for a Growing Future with Data*

Group 5:

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ABC
FOODMART

Introduction and Client Scenario



Overview of Client

ABC Foodmart is a growing local chain recently funded for significant expansion by investors

- 2 locations in Queens, expanding 3 in Brooklyn
- Investors request visibility into business metrics and profitability beyond operation



Challenge

Poor data management, leadership focused on operations

- Founder and CEO Mark focused on operating the chains over profitability
- Data silos lead to slow reporting, lagging analysis needed for strategic decisions



Objective

Transition manual data handling to a scalable, integrated relational database system

- Enable efficient operations, accurate profitability tracking and data-driven insights
- Design DBMS and create dashboards for management

Client Data Management Approach

 ABC_employees.xlsx

first_name	Employee first name
last_name	Employee last name
department	Working department
salary	Yearly salary (prorated)
hours_per_week	Contracted working hours
location_name	Store where employee works

Employee locations not mapped to unique store IDs

 ABC_products_and_suppliers.xlsx

productname	Name of the product
category	Product category (i.e., dairy)
unitcost	Cost per unit
unitprice	Retail price per unit
expiration_days	Expected shelf life in days
supplier_name	Supplying company name

Supplier and manufacturer info together; no separation of entities

 ABC_orders.xlsx

order_date	Date purchase order was placed
delivery_date	Delivery date
supplier_name	Supplier for the order
location	Store receiving the order
status	Status of the order
units	Units ordered

Inconsistent status values (i.e., 'shipped', 'ordered')

 ABC_sales_and_customers.xlsx

sale_date	Date of sale
location_name	Store location of the sale
cust_first_name	Customer first name
cust_last_name	Customer last name
product_name	Product sold
quantity	Quantity sold

Customers and sales info together

Operated with a strong focus on day-to-day operations over strategic analysis, leading to **disconnected flat files with no support for aggregated analysis**

Assessment of Data Challenges

Current Client Data Structure

ABC_sales_and_customers.xlsx

Illustrative

sale_date	location_name	customer_first	customer_last	age	gender	email	alt_email	product_name	quantity	unit_price	count_ap	motion	notes
2024-11-30 00:00:00	ABC Queen	James	Harris	43	O	amyvaughn	yes	Bottled Wa	2	1.21	0		
2024-12-04 00:00:00	ABC Queens	East						Bottled Wa	7	1.21	0		
2024-12-09 00:00:00	ABC Queen	Yvette	Gardner	42	F	robertmiller	yes	Bottled Wa	7	1.21	0		
2024-12-14 00:00:00	ABC Queens	East						Bottled Wa	2	1.21	0		
2024-12-14 00:00:00	ABC Queen	Angela	Archer	42	O	christinasr	yes	Bottled Wa	6	1.21	0		
2024-12-15 00:00:00	ABC Queen	Jeffrey	Jones	34	F	heidi35@e	yes	Bottled Wa	4	1.21	0		
2024-12-18 00:00:00	ABC Queen	David	Grimes	42	F	rgray@exa	no	Bottled Wa	8	1.21	0		
2024-12-29 00:00:00	ABC Queen	Sarah	Rowland	63	O	esparzajak	yes	Bottled Wa	8	1.21	0		
2025-01-04 00:00:00	ABC Queens	East						Bottled Wa	7	1.21	0		
2025-01-09 00:00:00	ABC Queen	Cindy	Edwards	25	M	ebrooks@e	no	Bottled Wa	5	1.21	0		
2025-01-12 00:00:00	ABC Queen	Alan	Flynn	81	M	brianna45	no	Bottled Wa	10	1.21	0		
2025-01-14 00:00:00	ABC Queens	East						Bottled Wa	7	1.21	0		
2025-01-18 00:00:00	ABC Queen	John	White	47	M	garzakelly	yes	Bottled Wa	10	1.21	0		
2025-01-31 00:00:00	ABC Queen	Katie	Daniels	66	M	john96@e	no	Bottled Wa	4	1.21	0		
2025-02-04 00:00:00	ABC Queen	Amber	Mcfarland	41	M	nicolerusse	yes	Bottled Wa	8	1.21	0		
2025-02-06 00:00:00	ABC Queen	James	Terry	72	M	michelle96	no	Bottled Wa	6	1.21	0		
2025-02-08 00:00:00	ABC Queen	Eric	Beck	51	O	diana67@e	yes	Bottled Wa	9	1.21	0		
2025-02-14 00:00:00	ABC Queen	Kristen	Rollins	32	M	amber19@	no	Bottled Wa	9	1.21	0		
2025-02-25 00:00:00	ABC Queens	East						Bottled Wa	3	1.21	0		
2025-03-01 00:00:00	ABC Queen	Amy	Harrington	81	O	mckeealan	no	Bottled Wa	2	1.21	0		
2025-03-02 00:00:00	ABC Queens	East						Bottled Wa	9	1.21	0		
2025-03-08 00:00:00	ABC Queen	Alexandra	Russo	26	M	phillip30@	yes	Bottled Wa	8	1.21	0		
2025-03-12 00:00:00	ABC Queen	Daniel	Johnson	83	O	patelsteve	yes	Bottled Wa	7	1.21	0		
2025-03-20 00:00:00	ABC Queen	Michele	Thompson	60	M	fwilliams@	no	Bottled Wa	3	1.21	0		
2025-03-21 00:00:00	ABC Queen	Kyle	Hall	39	M	vmills@ex	yes	Bottled Wa	8	1.21	0		
2025-03-26 00:00:00	ABC Queen	Janice	Lawson	33	O	edward53	yes	Bottled Wa	3	1.21	0		
2025-03-29 00:00:00	ABC Queens	East						Bottled Wa	8	1.21	0		
2025-03-30 00:00:00	ABC Queen	Ronald	Scott	41	M	ellisonhale	yes	Bottled Wa	7	1.21	0		

- 1 Critical return information is buried in a text field
- 2 Finding sales of different products requires manual searching and matching

Operational Impacts



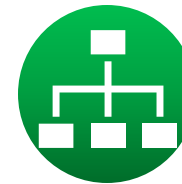
Data Inconsistency

- Inconsistent status values and varied naming conventions compromise data reliability



Lack of Integrated Analysis

- Combining data across multiple files is manual and prone to errors during analysis



Lack of Defined Relationships

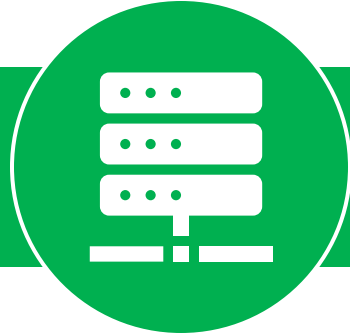
- Flat structure makes it hard to model one-to-many relationships

Team R&R Distribution



Database Architect

- Jaejun Lead
- Design logical and physical database schema
- Enforce normalization and referential integrity across system



ETL Developer

- Min Sung Lead
- Lead the development of Python-based ETL scripts
- Ensure smooth data migration to new database server



Data Analyst

- Nandiilkham Lead
- Create and optimize complex SQL queries to extract insights
- Address key business questions with actionable insights



BI & Reporting Lead

- Sara Lead
- Develop interactive dashboards using Metabase
- Ensure stakeholder alignment and data visualizations

Proposed Solution Architecture

Architecture Overview



Legacy Data Inputs

- Use client raw excel spreadsheets (4 in total) to input data



ETL Pipeline

- Data extraction, transformation and loading to new SQL database



Relational Database

- Normalized schema (of 16 tables) with referential integrity and automated triggers



End User Interactions and User Access

Data Analysts

- Raw data querying
- Data-driven insights
- View-access only



Executives

- Metabase dashboards
- Real-time KPIs
- High-level insights



New DBMS Key Benefits

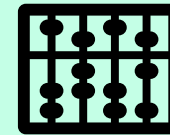
A scalable, secure data system empowering both strategic leadership and hands-on analysts with actionable insights



Single Source

All operational data is consolidated into one centralized relational database

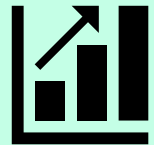
Eliminate Inconsistencies



Controlled Access

Strict role-based access for analysts (views), executives and HR/admins

Increased data security

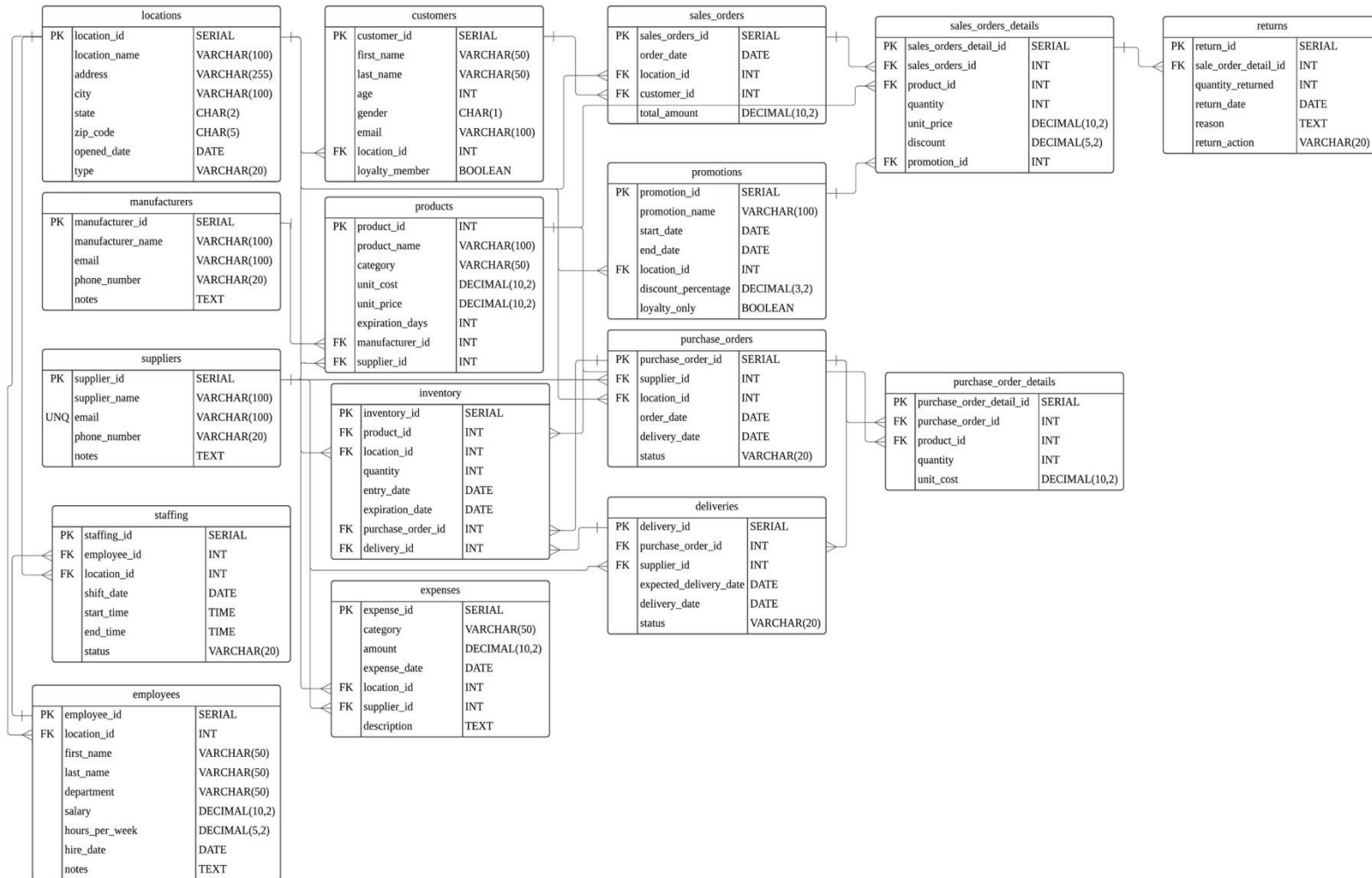


Data Analytics

Create custom-built procedures to empower business decisions

Derive actionable insights

Database Design Overview



Key Design Principles



Normalization (3NF)

- All tables were normalized to 3NF to eliminate redundancy and optimize data integrity



Referential Integrity

- PK and FK enforce valid relationships for products, suppliers, etc.



Operational Automation

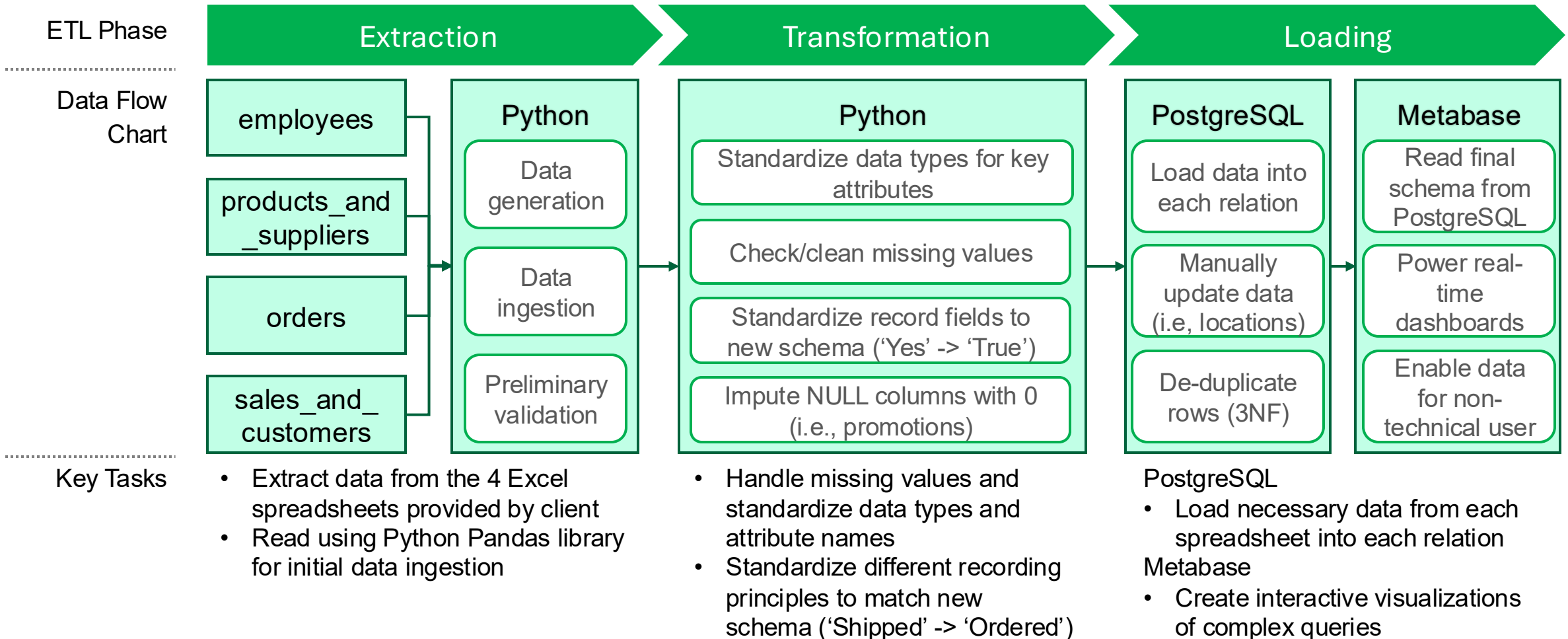
- 5 triggers to update inventory, delivery status, and total sales automatically



Scalability

- Designed to support multi-location expansion and higher transaction volumes

ETL Process Overview



[Back-Up] Client Data Mapping to Schema for Loading

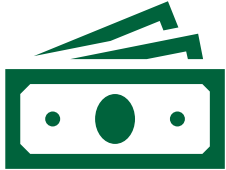
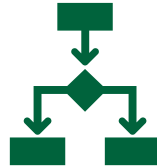
 ABC_sales_and_customers.xlsx

sale_date	sales_orders.order_date
location_name	Sales_orders.location_id
cust_first_name	dustomers.first_name
cust_last_name	customers.last_name
age	customers.age
gender	customers.gender
email	customers.email
loyalty_member	customers.loyalty_member
product_name	sales_orders_details.product_id
quantity	sales_orders_details.quantity
unit_price	sales_orders_details.unit_price
discount_applied	sales_orders_details.discount
promotion_name	promotions.promotion_id

 ABC_products_and_suppliers.xlsx

productname	products.product_name
category	products.category
unitcost	products.unit_cost
unitprice	products.unit_price
expiration_days	products.expiration_days
manufacturer_name	manufacturers.manufacturer_name
supplier_name	suppliers.supplier_name
supplier_email	suppliers.email
supplier_phone	suppliers.phone_number
notes	suppliers.notes

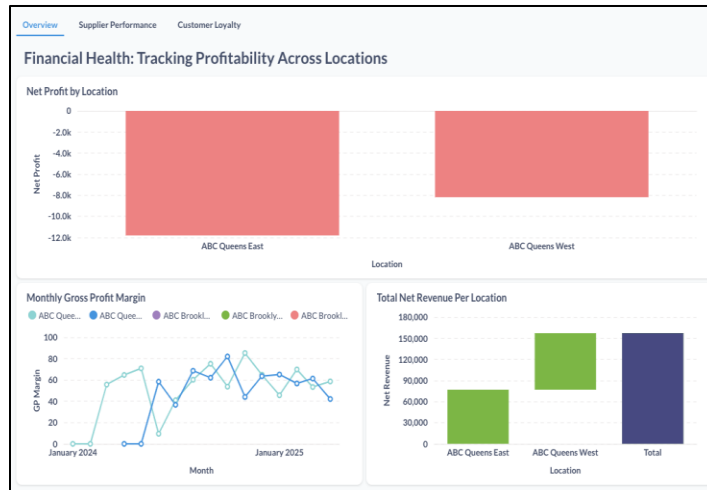
Analytical Procedures

Financial Performance	Sales Trends and Seasonality	Operational Improvements	Customer Loyalty
			
Understand store-level profitability and major cost • Net profit per location: measure profit/loss • Monthly gross margin: track COGS over time • COGS breakdown: identify high-cost products • Labor costs as a % of Net Revenue: evaluate employee cost	Optimize inventory, pricing and marketing • Top categories by net revenue: identify best-selling categories and prioritize promotions • Sales volume seasonality for key items: visualize seasonal demand shifts for products (like ice cream)	Improve supply chain reliability and reduce costs • Return rate of supplier products: quantify product quality issues via returns per supplier • Supplier rating: rank suppliers based on delivery punctuality and returns for renegotiation	Monitor engagement and retention from customers • Sales contribution for loyalty members: track revenue from members vs. guests • Customer repurchase rate: measure the rate of customers who return

Executive Dashboards and KPI Monitoring

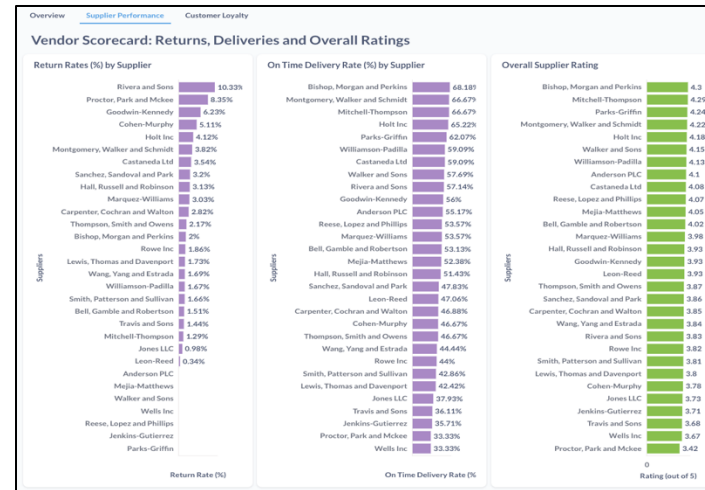
Implemented a 3-part Metabase dashboard for new leadership

Business Overview



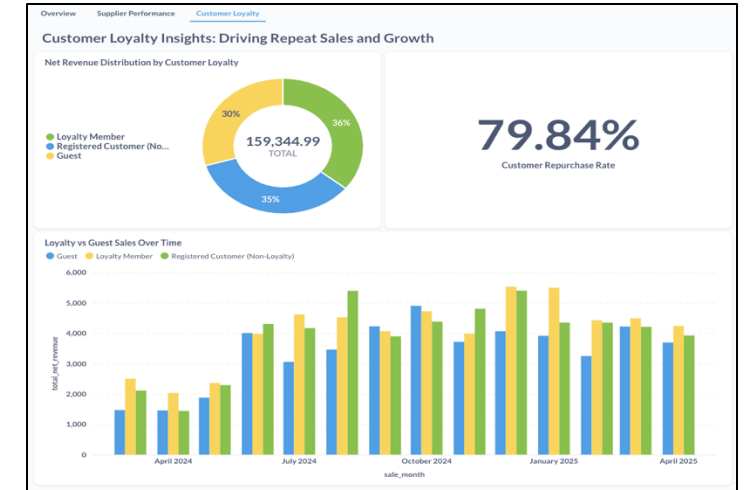
- Quick, top-down visibility into the health of each store
- Presents net revenue, gross profit, labor cost ratios, cost breakdowns by location

Supplier Reliability



- Visualization for return rates, on-time delivery percentages and supplier ratings
- Support risk management on factual performance indicators

Customer Loyalty



- Highlight revenue split between loyalty members and guests
- Enables quick overview as a baseline for customer lifetime value to evaluate ROI of loyalty initiatives

Key Findings and Strategic Impact

Category	 Financials	 Supplier Reliability	 Customer Loyalty
Key Insight	Near breakeven financials with net loss at ~10-15% • Outperforming early-stage retail benchmarks (20-30% losses)	Lost sales opportunities on missed on-time deliveries • Late deliveries likely contribute to stockouts and loss of sales	Strong customer loyalty between members • Outperformed industry benchmarks with 60% repurchase rate (+2-3x)
Strategic Impact	Strongly positioned to breakeven with minimal adjustments	Insights can enable renegotiation with underperforming vendors	High retention reduces customer attrition costs and creates sustainable revenue

Conclusions and Recommendations



Core Conclusions

- Real-time profitability and sales dashboards enable management to address margin pressure and staffing inefficiencies
- Supplier performance metrics now support data-backed renegotiation with vendors (and potential for consolidation)
- Loyalty program analytics identify high-value customers, enabling targeted retention strategies and personalized promotions



Future Recommendations

Category	Recommendations
Link Promotions to Product-Level ROI	Use promotions table to map campaigns to product-level sales to support campaign ROI tracking and future A/B testing
Loyalty Scoring Framework	Expand customer schema to differentiate loyalty members by behavior (frequency, basket size) to create targeted offers
Track Supplier Cost Variability	In addition to rating, capture unit cost volatility to factor in pricing stability and supply chain reliability for renegotiations



***Thank you for your
Attention!***
